



«Approved»  
Acting Dean of the School of Intelligent Systems  
**L.S. Aldasheva**  
2026 y.



## Syllabus Academic Year 2026 - 2027

| 1. General information                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |      |                 |
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| Course Code                                                    | CS 3145 KSC 3216                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |      |                 |
| Course Title                                                   | Computer Networks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |      |                 |
| Degree Cycle (Level)/<br>Major / Relation to<br>curriculum     | Bachelor's Degree<br>6B06301 – Cyber Security                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |      |                 |
| Year, trimester                                                | 2 <sup>nd</sup> year, 4 <sup>th</sup> trimester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |      |                 |
| Language of<br>Instruction:                                    | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |      |                 |
| Lecturer(s)/ Instructors/<br>Instructor Contact<br>Information | <p><b>Amirova Akzhibek</b>, PhD, Associate professor in the School of intelligent systems<br/> <a href="mailto:Akzhibek.amirova@astanait.edu.kz">Akzhibek.amirova@astanait.edu.kz</a><br/>           Contact details: Expo C1, Room C1.1.320</p> <p><b>Dossumbekov Yerlan</b>, PhD, Assistant Professor in the School of intelligent systems<br/> <a href="mailto:y.dossumbekov@astanait.edu.kz">y.dossumbekov@astanait.edu.kz</a><br/>           Contact details: Expo C1, Room C1.1.320</p> <p><b>Katrenova Zhanerke</b>, PhD, Assistant Professor in the School of intelligent systems<br/> <a href="mailto:Zh.katrenova@astanait.edu.kz">Zh.katrenova@astanait.edu.kz</a><br/>           Contact details: Expo C1, Room C1.1.320</p> <p><b>Yessirkepov Sharafatdin</b>, PhD, Assistant Professor in the School of intelligent systems<br/> <a href="mailto:Sharafatdin.yessirkepov@astanait.edu.kz">Sharafatdin.yessirkepov@astanait.edu.kz</a><br/>           Contact details: Expo C1, Room C1.1.320</p> <p><b>Kazhybayev Olzhas</b>, MSc, Senior lecturer in the School of intelligent systems<br/> <a href="mailto:o.kazhybayev@astanait.edu.kz">o.kazhybayev@astanait.edu.kz</a><br/>           Contact details: Expo C1, Room C1.1.320</p> <p><b>Satybaldiyeva Zhaina</b>, MSc, Instructor in the School of intelligent systems<br/> <a href="mailto:Zh.Satybaldiyeva@astanait.edu.kz">Zh.Satybaldiyeva@astanait.edu.kz</a><br/>           Contact details: Expo C1, Room C1.1.320</p> <p><b>Kambar Dinara</b>, MSc, Senior-lecturer in the School of intelligent systems<br/> <a href="mailto:d.kambar@astanait.edu.kz">d.kambar@astanait.edu.kz</a><br/>           Contact details: Expo C1, Room C1.1.320</p> <p><b>Baidullayeva Aigerim</b>, Instructor in the School of intelligent systems<br/> <a href="mailto:a.baidullaveva@astanait.edu.kz">a.baidullaveva@astanait.edu.kz</a><br/>           Contact details: Expo C1, Room C1.1.320</p> |               |      |                 |
| Number of Credits                                              | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |      |                 |
| Workload of<br>course components and                           | ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Contact hours | IASS | SIS Total hours |

|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |                   |                 |    |    |     |
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| credits per trimester                                           | credits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Lectures | Practice sessions | Laboratory work |    |    |     |
|                                                                 | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 20       | 30                | 0               | 10 | 90 | 150 |
| Prerequisites                                                   | Information and communications technology (ICT)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |                   |                 |    |    |     |
| Post requisites                                                 | Writing Diploma Work (Project) and Defence / Comprehensive Examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |                   |                 |    |    |     |
| <b>2. Goals, objectives and learning outcomes of the course</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |                   |                 |    |    |     |
| 1. Course Overview/Description                                  | “Computer Networks” is a 10-week course, where students will learn both the practical and conceptual skills to design and analyze computer communication networks. The course will cover basic and in-depth the most important concepts in contemporary networking and prepare students to plan and implement a network. Students will have hands-on experience using Cisco Packet Tracer network simulator tool by practicing every single chapter on labs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |                   |                 |    |    |     |
| 2. Course Learning Goals                                        | Course goal is to introduce students with the fundamental building blocks that form a modern network, such as protocols, topologies, hardware, and network operating systems. Moreover, to provide in-depth coverage of the most important concepts in contemporary networking, such as TCP/IP, Ethernet, wireless transmission, and security. Finally, to prepare students to pass Cisco certification exam (CCNA).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                   |                 |    |    |     |
| 3. Course Learning Outcomes                                     | Students will have a good understanding of the main elements that work together to form the Internet; will get familiar with diagnostic tools, network application development, and typical network designs and configurations.<br>Students will be able to build LANs, build small and medium scale network topologies, perform configurations for routers and switches, implement IP addressing schemes and understand Network automation tools and latest Cisco products.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |                   |                 |    |    |     |
| 4. Methods/forms of teaching                                    | The course is taught through a combination of lectures, practical sessions, laboratory work with Cisco Packet Tracer, Instructor-supervised and independent study, continuous assessments, and examinations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |                   |                 |    |    |     |
| 5. Coursework and Grading Scheme                                | Student achievement is assessed continuously and summatively: 60% of the grade comes from continuous assessment (weekly assignments, laboratory work defense, in-class activities, quizzes, and independent study tasks), distributed equally between the first attestation (30%) and the second attestation (30%); the final exam accounts for 40% of the overall grade. Each weekly assignment is graded on a 20-point scale, while the final exam combines theoretical (MCQ) and practical tasks. Grades are reported according to the university’s percentage-to-letter GPA conversion system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                   |                 |    |    |     |
| 6. Academic Integrity                                           | Academic integrity is a fundamental principle of this course. Students are expected to complete all assignments, laboratory work, quizzes, and examinations honestly and independently. Cheating, plagiarism, falsification of data, or submitting the same work for credit in multiple courses without prior consent are strictly prohibited.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |                   |                 |    |    |     |
| 7. Learning resources:                                          | <p>Main:</p> <ol style="list-style-type: none"> <li>1. Computer Networks: A Systems Approach / L. P. Larry, S. D. Bruce. - 6th ed. - USA Morgan Kaufmann, 2021. - 817 p. - ISBN 978-0-12-818200-0;</li> <li>2. Mayers Mike. CompTIA A+ Certification: All-in-One Exam Guide / M. Mayers, S. Jernigan. - 10 ed. - San Francisco: McGraw-Hill Education, 2019. - 1524 p. - ISBN 978-1-260-45403-1: 25500.00. 004 - K64;</li> <li>3. Lecture slides (available on moodle.astanait.edu.kz);</li> <li>4. Cisco Networking Academy Program CCNA 1 and 2 Companion Guide; CCNA 200-301 Official Cert Guide Volume 1.</li> <li>5. Computer Networks, Global Edition 5th Edition 2023- Andrew Tanenbaum, David Wetherall;</li> <li>6. Computer Networking: A Top-Down Approach / F.K. James, W.R. Keith. - 8 ed. - UK: Pearson, 2022. - 794 p. - ISBN 978-1-292-40546-5;</li> </ol> <p>Additional:</p> <ol style="list-style-type: none"> <li>1. Sadvakassova Assemgul. Methodological Guidelines for laboratory work in the discipline of Computer Networks: guideline / A. Sadvakassova, G. Tulepova. - Astana: AITU, 2024. - 79 p;</li> <li>2. Fundamentals of computer networks / S. Matthew, M.A. Gajetan. - USA: Humana Press, 2022. - 192 p. - ISBN 978-3-031-09416-3;</li> </ol> |          |                   |                 |    |    |     |



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| 8. Technology employed:    | <p>The course employs a blended technological environment that combines Moodle (moodle.astanait.edu.kz) as the primary Learning Management System for announcements, assignment submissions, quizzes, and exams; Microsoft Teams for synchronous online sessions, consultations, and webinars when in-person instruction is not possible; and Cisco Packet Tracer as the main simulation software for practicing network configuration, troubleshooting, and design. In addition, students will use Cisco Networking Academy resources, online journals, and professional articles to support independent study and enhance theoretical knowledge. In accordance with institutional teaching standards, students are expected to have consistent access to internet-enabled devices to engage effectively with LMS, communication platforms, and simulation tools.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 9. <u>Course Policies</u>  | <p><u>Course Policies for In-Person Instruction:</u></p> <ul style="list-style-type: none"> <li>- <u>Assignments Policy</u><br/>The course includes a structured List of Assignments, outlining weekly topics and tasks. Students will be assigned laboratory work at one of three complexity levels: Easy, Medium, or Hard. Each level is associated with a distinct point value, reflecting its degree of difficulty.</li> <li>- <u>Presence and Participation</u><br/>Students are expected to attend all scheduled class sessions with the required readings and supplementary materials completed beforehand. Attendance itself does not provide extra points, but it is mandatory for course completion. Students must participate in at least 70% of all class sessions; otherwise, they will not be admitted to the exam and will fail the course. Active participation is expected and will be evaluated by the quality of contributions, such as offering unique perspectives, advancing discussion, and building on peers' comments. Being more than 15 minutes late results in being marked absent.</li> <li>- <u>Deadlines and Late Submissions</u><br/>All assignments and homework must be submitted on time. Since assignments are often discussed in class on the due date, late submissions will not be accepted. Submissions through Moodle after the deadline will automatically reduce the overall score by 20 points, while late defenses of assignments result in 0%. In exceptional circumstances, students may request an extension in advance with proper justification; if approved, a new deadline will be assigned.</li> <li>- <u>The Honor Code of Students</u><br/>Cheating, plagiarism, and other forms of academic dishonesty are strictly prohibited. Examples include submitting work that is not your own, copying from sources without citation, falsifying data, or submitting the same work in two courses without consent. Any violation will result in a grade of 0 for the assignment and may lead to additional disciplinary measures under the University's Academic Conduct Policies.</li> <li>- <u>Course Policies for Use of Generative AI</u><br/>Generative AI tools (e.g., ChatGPT, Copilot) may only be used when explicitly allowed by the instructor. If permitted, students must clearly acknowledge the use of such tools and provide proper citations where relevant. Using AI to generate uncited content or to complete assignments dishonestly is considered plagiarism and will be subject to disciplinary measures.</li> <li>- <u>Extended Absences</u><br/>Students unable to attend class for extended periods must inform the instructor in advance and are responsible for covering missed material independently. Extended absences without a valid reason may prevent successful course completion. If more than 30% of classes are missed without excuse, the student will be marked "not graded" and denied access to the final exam.</li> <li>- <u>Late Homework Policy</u><br/>Late homework is generally not accepted. Work submitted after the deadline in Moodle will receive a 20-point deduction, and late defenses earn 0%. Exceptions are only made for extraordinary events, with prior approval and evidence, in which case a new deadline may be assigned.</li> <li>- <u>Exam Policy</u><br/>Two mid-term and end-term exams, along with a final exam, will be held. Exams include both theoretical (multiple-choice) and practical parts. Attendance and participation requirements must be met to qualify for exams. Use of unauthorized materials, electronic devices, or cheating during exams is strictly prohibited and will result in disciplinary action.</li> <li>- <u>Course Privacy Statement</u><br/>Course materials, lecture slides, assignments, and exams are provided for personal academic use only. Recording or sharing course content without instructor approval is prohibited. Student contributions in class are treated as part of a private academic setting and should not be disclosed outside the course environment without consent.</li> </ul> |
| 10. <u>Course Schedule</u> | <p>The course Computer Networks is designed as a 10-week program consisting of lectures, practice sessions, Instructor-supervised independent study (TSIS), and student independent study (SIS), with a total workload of 150 hours (5 ECTS). Each week covers both theoretical</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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|                                          | <p>concepts and practical applications, supported by independent assignments. Topics progress from fundamentals of networking to advanced issues of network security:</p> <p>Week 1 – Network Fundamentals, Basic Switch and End Device Configuration</p> <p>Week 2 – Protocols and Models</p> <p>Week 3 – Physical Layer and Number Systems</p> <p>Week 4 – Data Link Layer and Ethernet Switching</p> <p>Week 5 – Network Layer, Address Resolution, Basic Router Configuration</p> <p>Week 6 – IPv4 Addressing</p> <p>Week 7 – IPv6 Addressing and ICMP</p> <p>Week 8 – Transport Layer</p> <p>Week 9 – Application Layer</p> <p>Week 10 – Network Security Fundamentals</p> <p>Each week includes 2 lecture hours, 3 practice hours, 1 TSIS hour, and 9 SIS hours, with independent assignments focusing on lab-based tasks such as DHCP, NAT, ARP, DNS, troubleshooting, and network testing utilities. Continuous assessment is conducted through weekly assignments, mid-term and end-term evaluations, followed by a final exam</p> |
| 11. <u>Syllabus Inclusion Statements</u> | <p>This course is committed to fostering an inclusive and equitable learning environment that values diversity, respect, and collaboration. Students from all backgrounds, perspectives, and experiences are welcomed and encouraged to contribute to class discussions and activities. Course materials and assignments are designed to be accessible, and reasonable accommodations will be provided in accordance with university policies to ensure that all students can participate fully. The course promotes respect for diverse ideas and experiences, encourages critical engagement with different viewpoints, and emphasizes academic integrity and professional conduct. Students are also expected to respect the privacy of class discussions and materials, and to treat their peers and instructors with courtesy and professionalism at all times</p>                                                                                                                                                                     |

### 3.1 Abbreviations

| # | Abbreviation | Meaning                                                                                                                                                                                                                                                                        |
|---|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | IASS         | Instructor assisted student study - Instructors assist students in achieving success by delivering support, feedback, and resources. This support might take the form of clarifying difficult topics, recommending additional materials, or providing individualized guidance. |
| 2 | SIS          | Student independent study                                                                                                                                                                                                                                                      |
| 3 | IW           | Individual work                                                                                                                                                                                                                                                                |
| 4 | GW           | Group work                                                                                                                                                                                                                                                                     |
| 5 | PA           | Practical assignment                                                                                                                                                                                                                                                           |
| 6 | LW           | Laboratory work                                                                                                                                                                                                                                                                |
| 7 | MCQ          | Multiple choice quiz                                                                                                                                                                                                                                                           |
| 8 | PBT          | Paper-based test                                                                                                                                                                                                                                                               |

*NB: You may introduce abbreviations that are used in the syllabus*

### 3.2 Course Plan

| Week No | Course Topic                                                   | Lectures (h/w) | Practice sessions (h/w) | Lab. Sessions (h/w) | IASS (h/w) | SIS (h/w) |
|---------|----------------------------------------------------------------|----------------|-------------------------|---------------------|------------|-----------|
| 1       | Network fundamental. Basic Switch and End Device Configuration | 2              | 3                       | 0                   | 1          | 9         |
| 2       | Protocols and Models                                           | 2              | 3                       | 0                   | 1          | 9         |
| 3       | Physical Layer. Number Systems.                                | 2              | 3                       | 0                   | 1          | 9         |
| 4       | Data Link Layer. Ethernet Switching                            | 2              | 3                       | 0                   | 1          | 9         |



|    |                                                               |           |           |          |           |           |
|----|---------------------------------------------------------------|-----------|-----------|----------|-----------|-----------|
| 5  | Network Layer. Address Resolution. Basic Router Configuration | 2         | 3         | 0        | 1         | 9         |
| 6  | IPv4 Addressing                                               | 2         | 3         | 0        | 1         | 9         |
| 7  | IPv6 Addressing. ICMP                                         | 2         | 3         | 0        | 1         | 9         |
| 8  | Transport Layer                                               | 2         | 3         | 0        | 1         | 9         |
| 9  | Application Layer                                             | 2         | 3         | 0        | 1         | 9         |
| 10 | Network Security Fundamentals                                 | 2         | 3         | 0        | 1         | 9         |
|    | <b>Total:</b>                                                 | <b>20</b> | <b>20</b> | <b>0</b> | <b>10</b> | <b>90</b> |

### 3.3 Detailed course plan

| Week No | Lecture Topics with Learning Resources                                                                                                                                       | Tasks and Activities for Practice Sessions with Recommended Reading Resources                                                                                              | Lab Exercises, Assigned Activities, and Supporting Reading Resources (Some courses do not include lab work; in such cases, this column can be omitted) |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | <b>Week 1: Network Fundamentals, Basic Switch and End Device Configuration</b><br><i>Computer Networks (Tanenbaum &amp; Wetherall, 2023), CCNA Official Cert Guide Vol.1</i> | Basic Switch and End Device Configuration ( <i>Cisco CCNA Labs, Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ).                                                   | -                                                                                                                                                      |
| 2       | <b>Week 2: Protocols and Models – Computer Networking: A Top-Down Approach</b> (James & Keith, 2022)                                                                         | Investigate the TCP-IP and OSI Models in Action<br>Use Wireshark to View Network Traffic<br>( <i>Cisco CCNA Labs, Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ). | -                                                                                                                                                      |
| 3       | <b>Week 3: Physical Layer and Number Systems – Fundamentals of Computer Networks</b> (Matthew & Gajetan, 2022)                                                               | Connect a Wired and Wireless LAN<br>Connect the Physical Layer<br>( <i>Cisco CCNA Labs, Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ).                           | -                                                                                                                                                      |
| 4       | <b>Week 4: Data Link Layer and Ethernet Switching – Computer Networks: A Systems Approach</b> (Larry & Bruce, 2021)                                                          | Use Wireshark to Examine Ethernet Frames<br>( <i>Cisco CCNA Labs, Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ).                                                 | -                                                                                                                                                      |
| 5       | <b>Week 5: Network Layer, Address Resolution, Basic Router Configuration – Cisco Networking Academy CCNA 1 &amp; 2 Companion Guide</b>                                       | Troubleshoot Default Gateway Issues<br>Basic Device Configuration<br>( <i>Cisco CCNA Labs, Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ).                        | -                                                                                                                                                      |
| 6       | <b>Week 6: IPv4 Addressing – CompTIA A+ Certification All-in-One Exam Guide</b> (Mayers & Jernigan, 2019)                                                                    | Subnet an IPv4 Network<br>VLSM Design and Implementation Practice<br>( <i>Cisco CCNA Labs, Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ).                        | -                                                                                                                                                      |
| 7       | <b>Week 7: IPv6 Addressing and ICMP – Computer Networks</b> (Tanenbaum & Wetherall, 2023)                                                                                    | Configure IPv6 Addressing<br>( <i>Cisco CCNA Labs, Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ).                                                                | -                                                                                                                                                      |

|    |                                                                                                        |                                                                                                          |   |
|----|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---|
| 8  | <b>Week 8: Transport Layer</b> – <i>Computer Networking: A Top-Down Approach</i> (James & Keith, 2022) | TCP and UDP Communications (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ). | - |
| 9  | <b>Week 9: Application Layer</b> – <i>Computer Networks: A Systems Approach</i> (Larry & Bruce, 2021)  | Observe DNS Resolution (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ).     | - |
| 10 | <b>Week 10: Network Security Fundamentals</b> – Cisco Networking Academy online resources              | Secure Network Devices (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ).     | - |

### 3.4 List of assignments for Student Independent Study

| №  | Assignments (topics) for Independent Study     | Hours | Recommended literature and other sources (links)                              | Form of submission and deadlines |
|----|------------------------------------------------|-------|-------------------------------------------------------------------------------|----------------------------------|
| 1  | 2                                              | 3     | 4                                                                             | 5                                |
| 1  | Wireless Standards.<br>Set Up a Home Router    | 9     | (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ). | Labs                             |
| 2  | The IPv4 Address Structure.                    | 9     | (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ). | Labs                             |
| 3  | Dynamic Addressing with DHCP.                  | 9     | (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ). | Labs                             |
| 4  | Network Address Translation (NAT)              | 9     | (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ). | Labs                             |
| 5  | The Address Resolution Protocol (ARP) Process  | 9     | (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ). | Labs                             |
| 6  | Local and Remote Network Segments              | 9     | (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ). | Labs                             |
| 7  | Domain Name System.<br>Web Clients and Servers | 9     | (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ). | Labs                             |
| 8  | Application Layer Services.                    | 9     | (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ). | Labs                             |
| 9  | Troubleshooting Commands.                      | 9     | (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp; Wetherall, 2023</i> ). | Labs                             |
| 10 | Network Testing Utilities Summary.             | 9     | (Cisco CCNA Labs, <i>Computer Networks Tanenbaum &amp;</i>                    | Labs                             |

|  |  |  |                   |  |
|--|--|--|-------------------|--|
|  |  |  | Wetherall, 2023). |  |
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**3.5 For instructor assisted student study (IASS) schedule and policy follow your course LMS Moodle.**

**4. Student performance evaluation system for the course.**

| Period                      | Assignments*                                                          | Number of points | Form of assessment | Total      |
|-----------------------------|-----------------------------------------------------------------------|------------------|--------------------|------------|
| 1 <sup>st</sup> attestation | Assignment 1 (week 1)                                                 | 20               |                    | 100        |
|                             | Assignment 2 (week 2)                                                 | 20               |                    |            |
|                             | Assignment 3 (week 3)                                                 | 20               |                    |            |
|                             | Assignment 4 (week 4)                                                 | 20               |                    |            |
|                             | Midterm (week 5)                                                      | 20               | Paper based        |            |
| 2 <sup>nd</sup> attestation | Assignment 5 (week 6)                                                 | 15               |                    | 100        |
|                             | Assignment 6 (week 7)                                                 | 15               |                    |            |
|                             | Assignment 7 (week 8)                                                 | 15               |                    |            |
|                             | Assignment 8 (week 9)                                                 | 15               |                    |            |
|                             | Cisco certificate                                                     | 10               |                    |            |
|                             | Endterm (week 10)                                                     | 30               | Paper based        |            |
| Final exam                  | Paper based Exam (MCQ, open questions)                                |                  |                    | 100        |
| <b>Total</b>                | <b>0.3*1<sup>st</sup> Att + 0.3 *2<sup>nd</sup> Att + 0,4 * Final</b> |                  |                    | <b>100</b> |

\* The complete title of the assignment must be clearly stated.

\*\* The number of assignments may vary depending on the course content and is determined by the course syllabus.

\*\*\* Description of iROS or Social GPA elements, if applicable

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

| Letter Grade | GPA  | Percentage | Description  |
|--------------|------|------------|--------------|
| A            | 4,0  | 95-100     | Excellent    |
| A-           | 3,67 | 90-94      |              |
| B+           | 3,33 | 85-89      | Good         |
| B            | 3,0  | 80-84      |              |
| B-           | 2,67 | 75-79      |              |
| C+           | 2,33 | 70-74      |              |
| C            | 2,0  | 65-69      | Satisfactory |
| C-           | 1,67 | 60-64      |              |
| D+           | 1,33 | 55-59      |              |
| D            | 1,0  | 50-54      |              |
| FX           | 0    | 30-49      | Fail         |
| F            | 0    | 0-29       |              |

#### Course Assessment Rubric for Student Performance Evaluation

| Learning Outcomes                       | Indicators                                         | Excellent (90–100)                   | Good (70–89)                      | Satisfactory (50–69)                | Fail (0–49)                   |
|-----------------------------------------|----------------------------------------------------|--------------------------------------|-----------------------------------|-------------------------------------|-------------------------------|
| 1. Demonstrate knowledge of fundamental | - Recalls and defines key networking terminology - | Accurately recalls and describes all | Explains most concepts correctly, | Recalls some concepts, explanations | Cannot recall or define basic |



|                                                                                                |                                                                                                                                 |                                                                                        |                                                                        |                                                                       |                                                          |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------|
| computer networking concepts (protocols, OSI/TCP-IP, topologies).                              | Describes OSI and TCP/IP models - Lists topologies and device roles                                                             | concepts; uses precise definitions and examples                                        | minor errors                                                           | vague                                                                 | concepts                                                 |
| 2. Comprehend and interpret addressing and protocol functions (IPv4/IPv6, routing, switching). | - Explains IPv4/IPv6 addressing principles - Classifies protocols by OSI layers - Distinguishes routing vs. switching           | Explains in detail with correct classification and distinctions                        | Explains with some inaccuracies in classification                      | Partial understanding, superficial explanation                        | Cannot explain or distinguish addressing/protocols       |
| 3. Apply subnetting and device configuration skills using Cisco Packet Tracer.                 | - Calculates subnetting and address plans - Configures routers/switches - Demonstrates troubleshooting                          | Performs subnetting flawlessly; configures independently; troubleshooting effective    | Minor errors in subnetting/configuration; resolves most issues         | Frequent mistakes; incomplete configurations; limited troubleshooting | Unable to apply subnetting or configure devices          |
| 4. Analyse network problems and propose solutions.                                             | - Analyses packet flows and error logs - Compares routing strategies - Identifies connectivity failures                         | Performs deep analysis; compares strategies; identifies causes comprehensively         | Provides reasonable analysis; identifies most issues                   | Superficial analysis; identifies only obvious issues                  | Cannot analyse or identify network problems              |
| 5. Synthesize and design small-scale network solutions.                                        | - Designs network topology for given case - Integrates addressing/routing/switching - Proposes justified improvements           | Designs coherent, scalable solutions; integrates all elements; strong justification    | Designs workable solutions; partial justification; limited integration | Designs basic solutions; weak justification; lacks integration        | Fails to design or propose viable solutions              |
| 6. Evaluate network performance and security mechanisms.                                       | - Assesses network metrics (latency, throughput) - Evaluates ACLs, firewalls, and policies - Defends design during oral defense | Provides critical evaluation; assesses metrics accurately; defends design convincingly | Provides reasonable evaluation; identifies pros/cons; partial defense  | Basic evaluation; weak evidence; limited defense                      | Cannot evaluate performance or defend security decisions |

## 5. Course assessment methodology

Assessment is conducted continuously throughout the course, comprising 60% continuous assessment throughout the semester and 40% summative assessment during the examination session, totaling 100%. Continuous assessment evaluates students' ongoing performance through attendance, laboratory work defense, in-class activities, course assignments as well as independent study (SIS).

- **IASS (Instructor assisted student study)** - Instructors assist students in achieving success by delivering support, feedback, and resources. This support might take the form of clarifying difficult topics, recommending additional materials, or providing individualized guidance.
- **SIS** – Student engagement in self-directed study on assigned topics and the completion of relevant assignments given in the syllabus.
- **Mid-term and End-term** are a review of the topics covered and an assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- Final assessment for the course "*Computer Networks*" includes a **comprehensive final exam**, which consists of two parts: (1) a multiple-choice theoretical test covering the fundamental concepts, models, and principles studied during the course, and (2) a practical problem-solving section where students are expected to apply configuration and troubleshooting skills in simulated network scenarios. According to the course exam policy, students must attend at least **70% of the classes** to be admitted to the final assessment. The use of unauthorized materials, communication devices, or any form of cheating during the exam is strictly prohibited and will result in a grade of **0 (zero)**, with further disciplinary actions possible.



in line with the University Academic Integrity Policy. The final exam accounts for **40% of the overall course grade**, while continuous assessment throughout the semester contributes the remaining 60%.

#### **6. Lecturer(s) signature**

| Full name         | Job title                | Date       | Signature |
|-------------------|--------------------------|------------|-----------|
| Amirova A.        | PhD, Associate professor | 29.08.2026 |           |
| Dossumbekov Y.    | PhD, Assistant Professor | 29.08.2026 |           |
| Katrenova Z.      | PhD, Assistant Professor | 29.08.2026 |           |
| Yessirkepov S.    | PhD, Assistant Professor | 29.08.2026 |           |
| Kazhybayev O.     | MSc, Senior lecturer     | 29.08.2026 |           |
| Satybaldiyeva Zh. | MSc, Instructor          | 29.08.2026 |           |
| Kambar D.         | MSc, Senior-lecturer     | 29.08.2026 |           |
| Baidullayeva A.   | Instructor               | 29.08.2026 |           |

